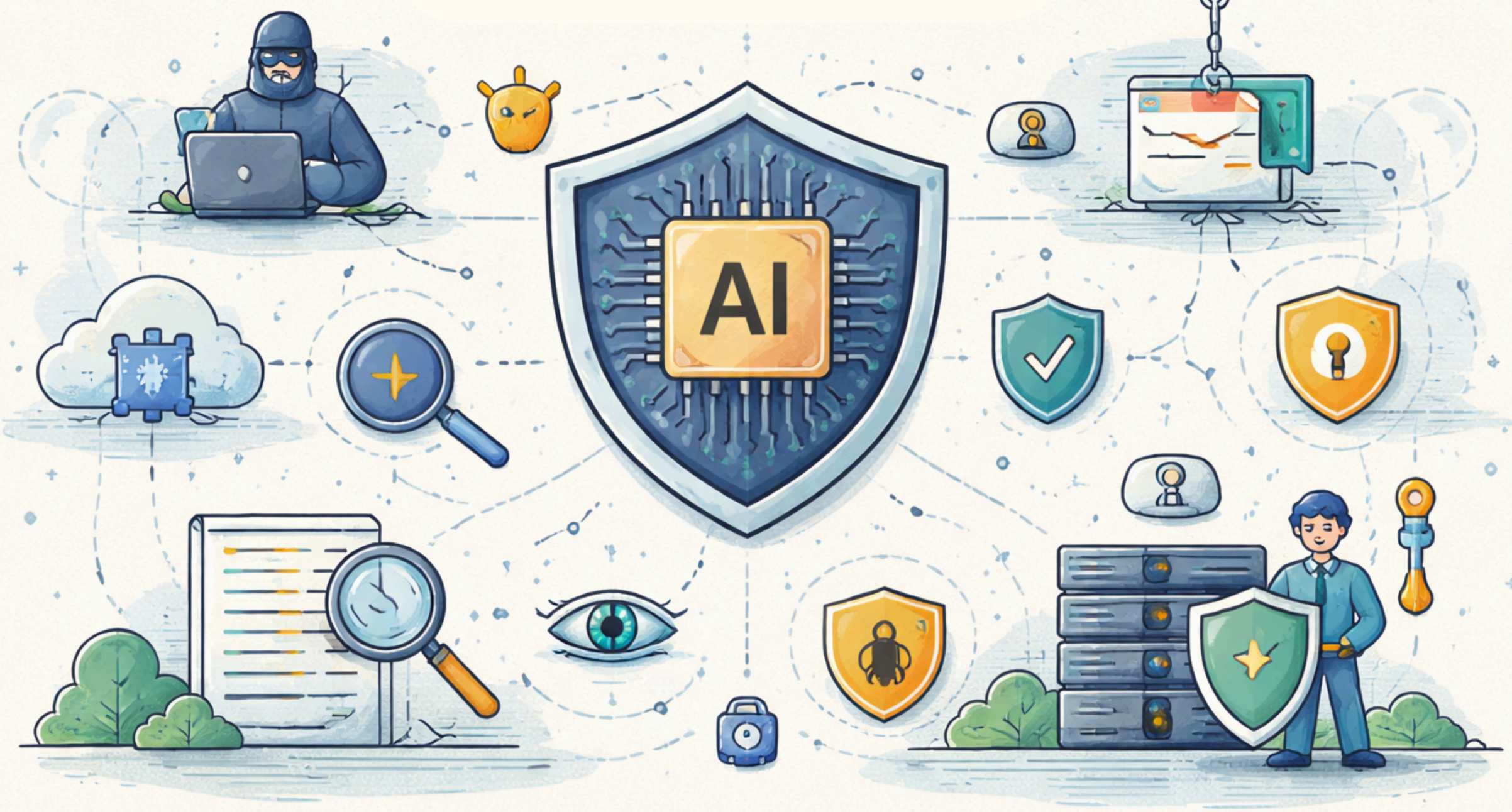

AI/LLM Security Architecture

New threats, New controls

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Threat 1

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Prompt Injection

Attack Vectors

- Direct jailbreaks override system prompts
- Indirect attacks via RAG poisoned documents
- Multi-turn context manipulation
- Delimiter confusion attacks

Mitigations

- Parameterized prompt templates
- Semantic input validation (not just syntax)
- Content classifiers before LLM processing
- Canary tokens in system prompts
- Input length limits per context window

Threat 2

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Data Exfiltration

Attack Vectors

- PII leakage in model outputs
- Training data extraction via memorization
- Distillation through synthetic queries
- Embedding space inference

Mitigations

- Output PII detection and redaction
- Differential privacy in training
- Response randomization techniques
- Prompt/response audit logs (immutable)
- Rate limiting per user + semantic clustering

Threat 3

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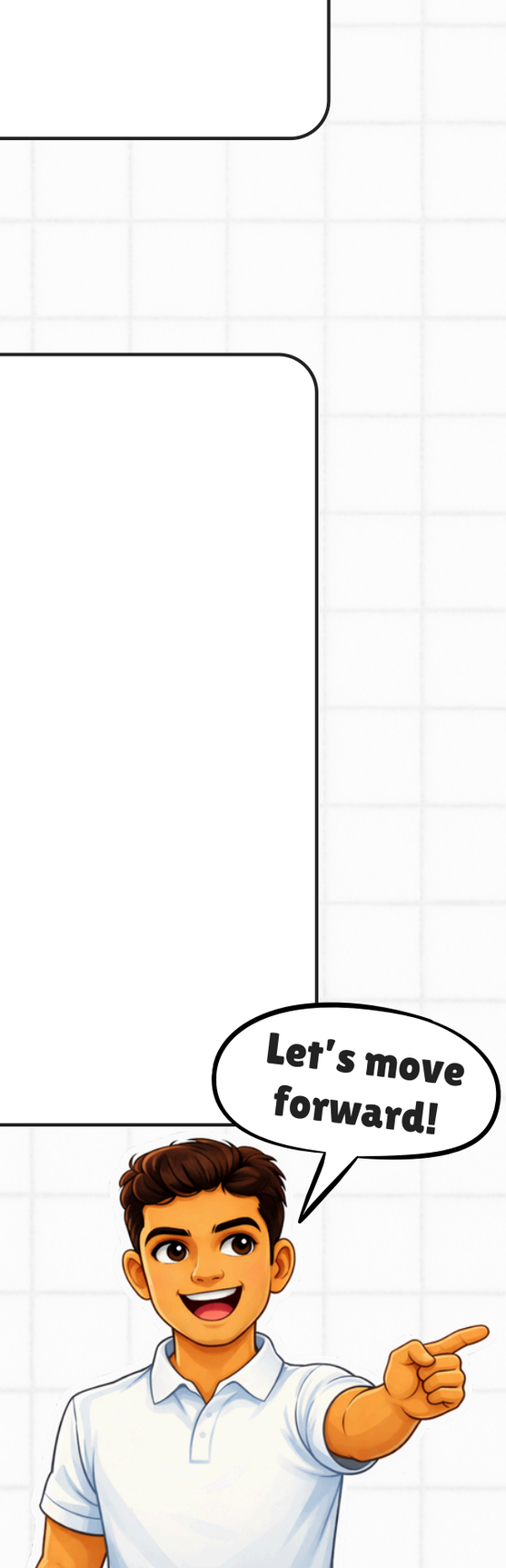
Model Theft

Attack Vectors

- Weight extraction via API timing analysis
- Model distillation attacks
- Fine-tuning dataset poisoning
- Supply chain compromise

Mitigations

- API response time randomization
- Query pattern anomaly detection
- Encrypted model storage
- Air-gapped fine-tuning environments
- Model access audit trails



Threat 4

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RAG Poisoning

Attack Vectors

- Malicious documents in vector database
- Citation manipulation
- Retrieval ranking exploits
- Embedding backdoors

Mitigations

- Document source authentication
- Vector DB access controls
- Retrieval result sandboxing
- Citation verification mechanisms
- Embedding model integrity validation

Threat 5

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Agent Exploits

Attack Vectors

- Tool-use privilege escalation
- Function calling with unsafe parameters
- Autonomous actions without approval
- Multi-agent coordination attacks

Mitigations

- Scoped tool permissions per agent
- Human-in-the-loop for critical actions
- Function call parameter validation
- Agent action audit logs
- State isolation between sessions

Threat 6

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Output Manipulation

Attack Vectors

- Hallucination exploitation
- Bias amplification attacks
- Toxicity injection
- Misinformation generation

Mitigations

- Secondary LLM output validation
- Hallucination detection scoring
- Toxicity and bias filters
- Watermarking generated content
- Output format constraints

Threat 7

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Training Data Poisoning

Attack Vectors

- Backdoor insertion in datasets
- Bias injection at scale
- Continuous learning contamination
- Adversarial examples in fine-tuning

Mitigations

- Dataset provenance tracking
- Automated bias detection
- Isolated fine-tuning pipelines
- Data validation before ingestion
- Version control for training datasets

Threat 8

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Compliance Violations

Attack Vectors

- ⚠ GDPR "right to be forgotten" conflicts
- ⚠ Unauditable model decisions
- ⚠ Copyright infringement in outputs
- ⚠ EU AI Act non-compliance

Mitigations

- ⚠ Model cards documenting lineage
- ⚠ Explainability mechanisms
- ⚠ Copyright detection in outputs
- ⚠ Risk classification per EU AI Act
- ⚠ Data retention policies for prompts

Threat 8

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Compliance Violations

Infrastructure Controls

ESSENTIAL DEFENSES

- Dedicated VPC for LLM workloads
- VPC endpoints for SageMaker/Bedrock
- Network egress monitoring
- GPU instance isolation
- Secrets rotation for API keys
- Immutable logging storage

Threat 8

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Compliance Violations

Monitoring & Detection

DETECTION MECHANISMS

- Anomaly detection in prompt patterns
- Token usage spike alerts
- Semantic drift monitoring
- Failed validation tracking
- Model performance degradation
- Abuse pattern clustering

Threat 8

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Compliance Violations

Cost-Effective Priorities

BUDGET ALLOCATION

- Input validation (highest ROI)
- Output filtering with secondary LLM
- Prompt/response logging
- Rate limiting + abuse detection
- Security spend: 10-15% of inference costs
- Red teaming quarterly assessments

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SOLUTION ARCHITECT

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